SUPPLEMENTARY MATERIAL: THE CERTIFICATE TABLES FOR “A DICHOTOMY FOR ERDŐS PROBLEM 1112”

JOHAN LAND

This supplement to the paper *A dichotomy for Erdős problem 1112: existence of* $r_k(d_1, d_2)$ *for* $k \geq 3$ lists in full the finite certificate tables referenced in its appendix on the certificate tables. Each entry is a hard-core triple (a, b, M) together with a box certificate (x, Y, Z), asserting that the multiset of x copies of a, Y of b and Z of M has budget $x + Y + Z \leq M - 1$ and subset sums containing M consecutive integers; see the main paper for the definitions, the two-generator/frame mechanism, and the reconciliation of printed rows against the rows decided by the Lean kernel.

These tables are generated mechanically from the canonical data (`data/certificate-data.md`) by `gen-tables.py`, which re-verifies every row — budget $\leq M - 1$ and coverage of M consecutive integers, by exact arbitrary-precision bitmask — at generation time, aborting on any failure. The same rows are transcribed into the Lean 4 development and decided by the kernel, and are exported in machine-readable form as `data/table-A.csv` and `data/table-B.csv` with SHA-256 digests in `data/SHA256SUMS`. Source, proof, and data: `github.com/beetree/math_erdos_1112`.

Table A: The 172 exceptional Case-T triples (the T-tail lemma of the main paper), all with $a \leq 29$; the scan over $a \leq 3000$ finds no others. Format $(a, b, M) \to (x, Y, Z)$: the multiset of x copies of a, Y of b, Z of M.

$(4,9,10)\to(7,1,1)$	$(4,13,14)\to(10,1,1)$	$(5,8,9)\to(5,2,1)$	$(5,9,12)\to(7,2,1)$
$(5,11,12)\to(7,1,2)$	$(5,11,13)\to(8,2,1)$	$(5,12,14)\to(9,1,2)$	$(5,13,14)\to(8,2,1)$
$(5,13,16)\to(10,1,2)$	$(6,11,14)\to(9,1,2)$	$(6,11,15)\to(9,2,1)$	$(6,13,14)\to(9,1,2)$
$(6,13,15)\to(10,2,1)$	$(6,13,16)\to(10,1,2)$	$(7,10,12)\to(7,3,1)$	$(7,10,15)\to(8,2,2)$
$(7,11,12)\to(7,2,2)$	$(7,11,13)\to(7,2,2)$	$(7,11,16)\to(8,3,1)$	$(7,12,13)\to(7,3,1)$
$(7,12,15)\to(8,2,2)$	$(7,13,17)\to(9,3,1)$	$(7,13,18)\to(9,2,2)$	$(7,15,16)\to(9,1,3)$
$(7,15,17)\to(9,2,2)$	$(7,15,18)\to(10,3,1)$	$(7,16,17)\to(10,2,2)$	$(7,16,18)\to(11,1,3)$
$(8,11,12)\to(7,3,1)$	$(8,11,15)\to(9,2,2)$	$(8,13,14)\to(7,3,2)$	$(8,13,15)\to(8,2,2)$
$(8,13,17)\to(10,2,2)$	$(8,15,18)\to(9,3,2)$	$(8,15,19)\to(11,2,2)$	$(8,17,18)\to(11,1,3)$
$(8,17,19)\to(10,2,2)$	$(9,13,16)\to(8,3,2)$	$(9,13,20)\to(9,4,1)$	$(9,14,15)\to(8,2,2)$
$(9,14,16)\to(9,4,1)$	$(9,14,17)\to(9,2,3)$	$(9,14,20)\to(9,3,2)$	$(9,16,17)\to(9,4,1)$
$(9,16,19)\to(10,3,2)$	$(9,17,20)\to(11,2,3)$	$(9,19,20)\to(11,1,4)$	$(10,13,15)\to(9,4,1)$
$(10,13,18)\to(10,3,2)$	$(10,13,21)\to(9,3,2)$	$(10,17,18)\to(9,3,2)$	$(10,17,19)\to(9,3,2)$
$(10,17,21)\to(10,2,3)$	$(11,14,15)\to(7,3,4)$	$(11,14,18)\to(9,5,1)$	$(11,14,21)\to(10,3,2)$
$(11,15,16)\to(7,3,2)$	$(11,15,20)\to(9,4,2)$	$(11,16,18)\to(8,4,2)$	$(11,16,19)\to(10,5,1)$
$(11,16,20)\to(9,3,2)$	$(11,17,18)\to(8,2,3)$	$(11,18,19)\to(9,4,3)$	$(11,18,20)\to(11,5,1)$
$(11,18,21)\to(10,2,3)$	$(11,19,20)\to(9,4,2)$	$(11,20,21)\to(11,5,1)$	$(12,17,18)\to(10,5,1)$
$(12,17,20)\to(10,3,2)$	$(12,17,21)\to(10,2,3)$	$(12,17,23)\to(12,4,2)$	$(12,19,20)\to(10,3,2)$
$(12,19,21)\to(11,2,3)$	$(12,19,22)\to(11,3,4)$	$(12,19,23)\to(11,4,2)$	$(13,16,21)\to(10,6,1)$
$(13,16,24)\to(11,2,4)$	$(13,17,20)\to(9,4,2)$	$(13,17,24)\to(10,4,4)$	$(13,18,19)\to(9,4,2)$
$(13,18,20)\to(9,3,3)$	$(13,18,22)\to(11,6,1)$	$(13,18,24)\to(11,3,3)$	$(13,19,24)\to(10,3,3)$
$(13,20,21)\to(10,2,4)$	$(13,20,22)\to(10,4,2)$	$(13,20,23)\to(12,6,1)$	$(13,20,24)\to(11,4,2)$
$(13,21,22)\to(10,4,4)$	$(13,21,24)\to(11,2,4)$	$(13,22,23)\to(11,3,3)$	$(13,22,24)\to(13,6,1)$
$(13,23,24)\to(11,4,2)$	$(14,17,24)\to(12,5,2)$	$(14,19,20)\to(9,3,3)$	$(14,19,21)\to(12,6,1)$
$(14,19,22)\to(9,5,2)$	$(14,19,25)\to(10,4,2)$	$(14,23,24)\to(12,3,5)$	$(14,23,25)\to(11,3,4)$

Date: July 5, 2026.

Table A: *(continued)*.

$(15,19,24)\to(11,2,4)$	$(15,22,24)\to(11,2,4)$	$(15,22,25)\to(11,4,2)$	$(15,22,26)\to(13,7,1)$
$(15,23,24)\to(11,2,4)$	$(15,23,26)\to(12,5,4)$	$(16,19,27)\to(13,6,2)$	$(16,21,23)\to(10,5,2)$
$(16,21,24)\to(13,7,1)$	$(16,21,25)\to(10,4,3)$	$(16,23,24)\to(13,7,1)$	$(16,23,26)\to(11,5,2)$
$(16,23,27)\to(12,3,4)$	$(16,25,26)\to(10,3,4)$	$(16,25,27)\to(12,2,5)$	$(17,21,28)\to(11,3,4)$
$(17,22,23)\to(10,5,6)$	$(17,22,25)\to(10,4,3)$	$(17,22,26)\to(10,6,2)$	$(17,22,28)\to(13,8,1)$
$(17,23,24)\to(10,3,4)$	$(17,24,25)\to(10,5,2)$	$(17,24,26)\to(11,4,4)$	$(17,24,28)\to(11,3,4)$
$(17,25,28)\to(11,6,2)$	$(17,26,27)\to(11,2,5)$	$(17,26,28)\to(11,4,3)$	$(17,27,28)\to(12,4,3)$
$(18,23,27)\to(14,8,1)$	$(18,23,29)\to(12,4,3)$	$(18,25,26)\to(10,5,3)$	$(18,25,27)\to(15,8,1)$
$(18,25,28)\to(11,3,4)$	$(19,24,30)\to(11,5,3)$	$(19,25,30)\to(11,4,3)$	$(19,26,27)\to(10,4,4)$
$(19,26,28)\to(11,3,4)$	$(19,26,29)\to(12,3,4)$	$(19,26,30)\to(12,6,2)$	$(19,27,28)\to(12,6,2)$
$(19,28,30)\to(13,3,5)$	$(19,29,30)\to(13,2,6)$	$(20,27,28)\to(11,3,4)$	$(20,27,29)\to(12,6,2)$
$(20,27,30)\to(16,9,1)$	$(20,27,31)\to(12,5,4)$	$(20,29,30)\to(16,9,1)$	$(21,29,30)\to(10,5,4)$
$(21,29,32)\to(12,3,5)$	$(22,29,31)\to(11,6,3)$	$(22,29,32)\to(11,5,4)$	$(22,29,33)\to(17,10,1)$
$(22,31,32)\to(12,5,3)$	$(22,31,33)\to(18,10,1)$	$(23,30,31)\to(13,7,8)$	$(23,30,33)\to(13,7,2)$
$(23,30,34)\to(13,4,4)$	$(23,31,32)\to(11,3,5)$	$(23,31,34)\to(11,6,4)$	$(23,32,33)\to(13,3,5)$
$(23,32,34)\to(12,4,4)$	$(23,33,34)\to(13,7,2)$	$(24,31,35)\to(12,4,4)$	$(25,33,36)\to(14,8,2)$
$(25,34,35)\to(13,4,4)$	$(25,34,36)\to(13,3,6)$	$(26,35,36)\to(13,3,6)$	$(26,35,37)\to(13,6,3)$
$(27,37,38)\to(14,6,3)$	$(28,37,39)\to(14,4,5)$	$(29,38,39)\to(15,9,10)$	$(29,39,40)\to(14,3,7)$

Table B: The 178 Case-B class bases ($a \leq 11$, $\mu \geq 12$). Format $[a;\bar{e},h]$ followed by the base triple $(a,b,M) \to (x,Y,Z)$. The λ-lift lemma of the main paper extends each certificate to every member of its class with larger e.

$[4;1,1]\,(4,17,18)\to(13,1,1)$	$[5;1,1]\,(5,16,17)\to(10,1,2)$	$[5;1,2]\,(5,16,18)\to(11,2,1)$
$[5;2,2]\,(5,17,19)\to(12,1,2)$	$[5;3,1]\,(5,18,19)\to(11,2,1)$	$[5;3,3]\,(5,18,21)\to(13,1,2)$
$[5;4,3]\,(5,14,17)\to(10,2,1)$	$[6;1,1]\,(6,19,20)\to(13,1,2)$	$[6;1,2]\,(6,19,21)\to(14,2,1)$
$[6;1,3]\,(6,19,22)\to(14,1,2)$	$[6;5,3]\,(6,17,20)\to(13,1,2)$	$[6;5,4]\,(6,17,21)\to(13,2,1)$
$[7;1,1]\,(7,22,23)\to(13,1,3)$	$[7;1,2]\,(7,22,24)\to(13,2,2)$	$[7;1,3]\,(7,22,25)\to(14,3,1)$
$[7;1,4]\,(7,15,19)\to(11,2,2)$	$[7;2,1]\,(7,23,24)\to(14,2,2)$	$[7;2,2]\,(7,23,25)\to(15,1,3)$
$[7;2,4]\,(7,16,20)\to(11,2,2)$	$[7;3,2]\,(7,17,19)\to(11,3,1)$	$[7;3,3]\,(7,17,20)\to(12,1,3)$
$[7;3,5]\,(7,17,22)\to(12,2,2)$	$[7;4,1]\,(7,18,19)\to(11,2,2)$	$[7;4,2]\,(7,18,20)\to(11,2,2)$
$[7;4,4]\,(7,18,22)\to(13,1,3)$	$[7;4,5]\,(7,18,23)\to(12,3,1)$	$[7;5,1]\,(7,19,20)\to(11,3,1)$
$[7;5,3]\,(7,19,22)\to(12,2,2)$	$[7;5,5]\,(7,19,24)\to(14,1,3)$	$[7;6,3]\,(7,20,23)\to(13,2,2)$
$[7;6,4]\,(7,20,24)\to(13,3,1)$	$[7;6,5]\,(7,20,25)\to(13,2,2)$	$[8;1,1]\,(8,25,26)\to(16,1,3)$
$[8;1,2]\,(8,25,27)\to(14,2,2)$	$[8;1,3]\,(8,17,20)\to(12,3,1)$	$[8;1,4]\,(8,17,21)\to(12,2,2)$
$[8;1,5]\,(8,17,22)\to(11,3,2)$	$[8;3,1]\,(8,19,20)\to(12,3,1)$	$[8;3,3]\,(8,19,22)\to(14,1,3)$
$[8;3,4]\,(8,19,23)\to(14,2,2)$	$[8;3,6]\,(8,19,25)\to(12,2,2)$	$[8;5,1]\,(8,21,22)\to(11,3,2)$
$[8;5,2]\,(8,21,23)\to(12,2,2)$	$[8;5,4]\,(8,21,25)\to(15,2,2)$	$[8;5,5]\,(8,21,26)\to(16,1,3)$
$[8;7,3]\,(8,23,26)\to(13,3,2)$	$[8;7,4]\,(8,23,27)\to(16,2,2)$	$[8;7,5]\,(8,15,20)\to(11,3,1)$
$[8;7,6]\,(8,15,21)\to(10,2,2)$	$[9;1,1]\,(9,28,29)\to(16,1,4)$	$[9;1,2]\,(9,19,21)\to(12,2,2)$
$[9;1,3]\,(9,19,22)\to(12,3,2)$	$[9;1,4]\,(9,19,23)\to(12,4,1)$	$[9;1,5]\,(9,19,24)\to(13,2,2)$
$[9;1,6]\,(9,19,25)\to(12,4,2)$	$[9;2,1]\,(9,20,21)\to(12,2,2)$	$[9;2,2]\,(9,20,22)\to(13,1,4)$
$[9;2,3]\,(9,20,23)\to(13,2,3)$	$[9;2,4]\,(9,20,24)\to(13,2,2)$	$[9;2,6]\,(9,20,26)\to(13,3,2)$
$[9;4,2]\,(9,22,24)\to(13,2,2)$	$[9;4,3]\,(9,22,25)\to(13,3,2)$	$[9;4,4]\,(9,22,26)\to(15,1,4)$
$[9;4,6]\,(9,22,28)\to(13,4,2)$	$[9;4,7]\,(9,22,29)\to(14,4,1)$	$[9;5,1]\,(9,23,24)\to(13,2,2)$
$[9;5,2]\,(9,23,25)\to(14,4,1)$	$[9;5,3]\,(9,23,26)\to(14,2,3)$	$[9;5,5]\,(9,23,28)\to(16,1,4)$
$[9;5,6]\,(9,23,29)\to(14,3,2)$	$[9;5,7]\,(9,14,21)\to(10,2,2)$	$[9;7,1]\,(9,25,26)\to(14,4,1)$
$[9;7,3]\,(9,25,28)\to(15,3,2)$	$[9;7,5]\,(9,16,21)\to(11,2,2)$	$[9;7,6]\,(9,16,22)\to(10,4,2)$
$[9;7,7]\,(9,25,32)\to(18,1,4)$	$[9;8,3]\,(9,26,29)\to(16,2,3)$	$[9;8,4]\,(9,17,21)\to(11,2,2)$
$[9;8,5]\,(9,17,22)\to(11,4,1)$	$[9;8,6]\,(9,17,23)\to(11,3,2)$	$[9;8,7]\,(9,17,24)\to(12,2,2)$
$[10;1,1]\,(10,21,22)\to(13,1,4)$	$[10;1,2]\,(10,21,23)\to(11,2,3)$	$[10;1,3]\,(10,21,24)\to(11,3,2)$

Table B: *(continued)*.

$[10;1,4]\,(10,21,25)\to(14,4,1)$	$[10;1,5]\,(10,21,26)\to(14,3,2)$	$[10;1,6]\,(10,21,27)\to(13,3,2)$
$[10;1,7]\,(10,21,28)\to(14,3,3)$	$[10;3,1]\,(10,23,24)\to(12,3,3)$	$[10;3,2]\,(10,23,25)\to(15,4,1)$
$[10;3,3]\,(10,23,26)\to(16,1,4)$	$[10;3,5]\,(10,23,28)\to(16,3,2)$	$[10;3,6]\,(10,23,29)\to(14,2,3)$
$[10;3,8]\,(10,23,31)\to(14,3,2)$	$[10;7,1]\,(10,27,28)\to(14,3,2)$	$[10;7,2]\,(10,27,29)\to(14,3,2)$
$[10;7,4]\,(10,27,31)\to(15,2,3)$	$[10;7,5]\,(10,17,22)\to(12,3,2)$	$[10;7,7]\,(10,27,34)\to(20,1,4)$
$[10;7,8]\,(10,17,25)\to(12,4,1)$	$[10;9,3]\,(10,19,22)\to(11,3,3)$	$[10;9,4]\,(10,19,23)\to(11,3,2)$
$[10;9,5]\,(10,19,24)\to(13,3,2)$	$[10;9,6]\,(10,19,25)\to(13,4,1)$	$[10;9,7]\,(10,19,26)\to(11,3,2)$
$[10;9,8]\,(10,19,27)\to(12,2,3)$	$[11;1,1]\,(11,23,24)\to(13,1,5)$	$[11;1,2]\,(11,23,25)\to(12,2,3)$
$[11;1,3]\,(11,23,26)\to(12,3,2)$	$[11;1,4]\,(11,23,27)\to(13,3,4)$	$[11;1,5]\,(11,23,28)\to(14,5,1)$
$[11;1,6]\,(11,23,29)\to(14,2,4)$	$[11;1,7]\,(11,23,30)\to(13,4,2)$	$[11;1,8]\,(11,23,31)\to(14,4,3)$
$[11;2,1]\,(11,24,25)\to(12,2,4)$	$[11;2,2]\,(11,24,26)\to(15,1,5)$	$[11;2,3]\,(11,24,27)\to(12,4,2)$
$[11;2,4]\,(11,24,28)\to(13,2,3)$	$[11;2,5]\,(11,24,29)\to(13,4,3)$	$[11;2,6]\,(11,24,30)\to(13,3,2)$
$[11;2,8]\,(11,24,32)\to(15,4,2)$	$[11;3,1]\,(11,25,26)\to(12,3,4)$	$[11;3,2]\,(11,25,27)\to(12,4,3)$
$[11;3,3]\,(11,25,28)\to(16,1,5)$	$[11;3,4]\,(11,25,29)\to(15,5,1)$	$[11;3,6]\,(11,25,31)\to(14,2,3)$
$[11;3,7]\,(11,25,32)\to(15,2,4)$	$[11;3,9]\,(11,14,23)\to(9,3,2)$	$[11;4,1]\,(11,26,27)\to(12,3,2)$
$[11;4,2]\,(11,26,28)\to(13,2,4)$	$[11;4,4]\,(11,26,30)\to(17,1,5)$	$[11;4,5]\,(11,26,31)\to(14,3,4)$
$[11;4,6]\,(11,26,32)\to(14,4,2)$	$[11;4,8]\,(11,15,23)\to(10,2,3)$	$[11;4,9]\,(11,15,24)\to(10,5,1)$
$[11;5,2]\,(11,27,29)\to(13,4,2)$	$[11;5,3]\,(11,27,30)\to(16,5,1)$	$[11;5,4]\,(11,27,31)\to(14,3,2)$
$[11;5,5]\,(11,27,32)\to(18,1,5)$	$[11;5,7]\,(11,16,23)\to(10,4,3)$	$[11;5,8]\,(11,16,24)\to(11,3,2)$
$[11;5,9]\,(11,16,25)\to(11,4,2)$	$[11;6,1]\,(11,28,29)\to(13,2,3)$	$[11;6,2]\,(11,28,30)\to(14,3,4)$
$[11;6,3]\,(11,28,31)\to(14,2,4)$	$[11;6,4]\,(11,28,32)\to(15,4,3)$	$[11;6,6]\,(11,28,34)\to(19,1,5)$
$[11;6,7]\,(11,17,24)\to(10,3,2)$	$[11;6,8]\,(11,17,25)\to(11,5,1)$	$[11;6,9]\,(11,17,26)\to(11,4,2)$
$[11;7,1]\,(11,29,30)\to(14,4,3)$	$[11;7,2]\,(11,29,31)\to(17,5,1)$	$[11;7,3]\,(11,29,32)\to(15,2,3)$
$[11;7,5]\,(11,18,23)\to(10,4,2)$	$[11;7,6]\,(11,18,24)\to(11,4,2)$	$[11;7,7]\,(11,29,36)\to(20,1,5)$
$[11;7,9]\,(11,18,27)\to(13,3,2)$	$[11;8,1]\,(11,30,31)\to(14,4,2)$	$[11;8,2]\,(11,30,32)\to(15,3,2)$
$[11;8,4]\,(11,19,23)\to(11,2,4)$	$[11;8,5]\,(11,19,24)\to(11,2,3)$	$[11;8,7]\,(11,19,26)\to(12,5,1)$
$[11;8,8]\,(11,30,38)\to(21,1,5)$	$[11;8,9]\,(11,19,28)\to(12,4,3)$	$[11;9,1]\,(11,31,32)\to(17,5,1)$
$[11;9,3]\,(11,20,23)\to(11,3,4)$	$[11;9,5]\,(11,20,25)\to(11,3,2)$	$[11;9,6]\,(11,20,26)\to(12,4,3)$
$[11;9,7]\,(11,20,27)\to(12,2,3)$	$[11;9,8]\,(11,20,28)\to(12,4,2)$	$[11;9,9]\,(11,31,40)\to(22,1,5)$
$[11;10,3]\,(11,21,24)\to(11,4,3)$	$[11;10,4]\,(11,21,25)\to(11,4,2)$	$[11;10,5]\,(11,21,26)\to(12,2,4)$
$[11;10,6]\,(11,21,27)\to(13,5,1)$	$[11;10,7]\,(11,21,28)\to(13,4,2)$	$[11;10,8]\,(11,21,29)\to(12,3,2)$
$[11;10,9]\,(11,21,30)\to(13,2,3)$		